Multi-Metric Study of Language in German Physics Teaching

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Abstract

Technical language in physics teaching is widely perceived as difficult and as a learning barrier, in large part because it forms a register distinct from everyday language along many dimensions. We characterize German physics-teaching language and its differences from a general-language corpus using 33 metrics of linguistic complexity and readability, asking in which respects German physics language is particularly difficult and how metrics agree on the answer. We find that general language is lexically more diverse, whereas physics concentrates on a narrow, cohesive terminological core — a difference that surface simplification may fail to address. Complexity also shifts with educational level rather than rising uniformly: at university, explanatory load moves from nominal style toward mathematical formalism. Paradidactic texts, the one register written for a broad audience, sit closest to the reference corpus in vocabulary yet are rated among the least readable by readability formulas, while Wikipedia is furthest from it. Throughout, we treat divergences between metrics as themselves informative, revealing the assumptions each carries when applied to a specialized register.

1 Introduction

To learn physics is, to some extent, to acquire a specialized language which has distinct characteristics from those of everyday language (Leisen, 2006). Learners acquire a general everyday language register prior to formal instruction and must acquire this technical language while they learn scientific concepts (Lemke, 1990; Schleppegrell, 2001). The acquisition of technical language is widely regarded as necessary for consolidating conceptual learning (Rincke, 2007) and also for the development of specific communicative skills demanded in school and technical contexts (Kulgemeyer, 2010).

The features of technical language in physics teaching and their influence on physics learning have been the focus of numerous studies, and corpus-based work has begun to quantify aspects of physics language (e.g. Vuković Stamatović, 2020; Strohmaier et al., 2023). This paper contributes a broad corpus analysis quantifying how the complexity and readability of physics technical language in German change across school levels and text types, and how they compare with general-language texts. It also asks what those metrics see and fail to see in text from a technical register, since simplification needs reliable means both to diagnose what makes a text hard and to verify that an intervention has helped.

Two research questions guide the analysis. RQ1: To what extent do readability and complexity metrics differentiate physics texts from general written German, and which metric families are most sensitive? RQ2: Do the metrics converge on a consistent characterization, or do their divergences reveal the assumptions each carries — and what follows for their use in text simplification? To address these questions, we use the TeCoPhy corpus (Lacerda Fontanella et al., 2023), a multi-level collection of German physics texts spanning middle school to university, together with a written general-language reference corpus built from the Leipzig Corpora Collection (Quasthoff et al., 2014) with newspaper and internet texts. We apply and compare multiple established readability and complexity metrics, interpreting their results in relation to known features of technical language.

2 Related Work

In lexical quantitative studies in the context of physics teaching, Merzyn (1994) found that the number of new terms per lesson in physics courses was very high and that textbooks contain many terms occurring only once or twice. Similarly, Yun (2020) found a significant terminological gap between a high-school and a uni-

versity physics textbook through network analysis. Hancke et al. (2012) showed that lexical, syntactic, and morphological features classify German corpora by difficulty (easy/difficult) with high accuracy. Bryant et al. (2017) tested whether linguistic complexity increases systematically across grade levels and differs between school tracks in German lower-secondary geography textbooks, and found the expected progression held only partially. Vuković Stamatović (2020) is the most similar work to our study, building a corpus of various physics genres in English and analyzing it with a focus mostly on vocabulary metrics. Fraser (2025) manipulated the linguistic form of ninth-grade physics texts and found that simplification — whether at the text surface or through deeper restructuring — produced no significant gain in student comprehension, which was instead dominated by readers' prior knowledge and vocabulary. This echoes broader findings (e.g. Strohmaier et al., 2023) that surface-level readability manipulation does not improve comprehension of texts in science, technology, engineering and mathematics (STEM). Unlike Fraser (2025), we use the metrics as dependent variables for characterizing the inherent linguistic profile of a corpus.

3 Methodology

3.1 Corpora

TeCoPhy is a purpose-built corpus of German physics teaching materials covering five subcategories: middle-school textbooks, high-school textbooks, university textbooks, German Wikipedia physics articles, and a paradidactic category of popular-science texts. A subset from the Leipzig Collection (namely the *deu-mixed-typical-2011*, *deu-web-public-2019* and *deu-news-2024*) (Quasthoff et al., 2014), primarily consisting of newspaper and internet texts, served as a general-language reference and is referred to hereafter as the Leipzig Corpus. Table 1 presents corpus sizes.

Table 1: Corpus description: sentences, tokens, and vocabulary types per corpus and TeCoPhy subcategory.

Corpus	Sentences	Tokens	Types
Leipzig	2,999,897	43,070,460	1,369,048
TeCoPhy (all)	778,195	19,117,869	343,069
Paradidactics	88,990	2,132,631	113,033
Middle school	75,264	1,255,276	59,668
High school	93,468	4,775,256	70,039
University	484,428	10,191,997	227,491
Wikipedia	36,045	762,709	54,172

The corpora were tokenized and lemmatized using SoMaJo (Proisl and Uhrig, 2016) and HanTa (Wartena, 2019). Digits, mathematical symbols and units were replaced by placeholders and each metric calculation could be configured to include or exclude these placeholders, according to what the measure requires. Lemmatized tokens were used for all metrics except readability formulas, compound proxy, symbol density, and average word length, which use raw surface tokens. For some metrics, we applied a sentence gate, excluding from the calculation sentences with more than 10% non-natural tokens, i.e. symbols and mathematical formulas.

3.2 Metrics

Thirty-three metrics were computed across five groups: surface and readability, lexical diversity, information-theoretic, syntactic and morphological, and cross-corpus distributional distance. Size-sensitive metrics used a sampling procedure (ten random samples of different fixed sizes, depending on the metric). Metrics requiring global distributional structure used a single sample truncated to the smallest category corpus (36,034 sentences; 778,943 tokens for the token-level measures). Distance metrics were computed pairwise against the Leipzig Corpus. In the Limitations section, we report which metrics separate consistently across draws.

3.2.1 Surface and Readability

Surface metrics are computed over the original tokens of the gated sentences, with the exception of symbol density. The Flesch–Amstad index (*F-Amstad*; Amstad, 1978) and the first Wiener Sachtextformel (*WSTF*; Bamberger and Vanecek, 1984) are established readability formulas for German texts. Symbol density (*SymDens*) is the proportion of tokens that are mathematical-expression placeholders or isolated single alphabetic characters functioning as mathematical symbols. We additionally report the distribution of sentence lengths — its mean (*Sent.(mean)*) and standard deviation (*Sent.(std)*) — together with mean word length, both as readability-related surface features and to help identify which surface variable most strongly drives the readability indices.

3.2.2 Lexical Diversity

Lexical diversity was quantified under two sampling strategies. In the *global* strategy, each

metric was computed over a single large, order-independent sample drawn from the corpus group, capturing thematic lexical breadth. In the *sequential* strategy, denoted *ord*, the metric was estimated by bootstrapping over contiguous chunks that preserve sentence order, so that it reflects local lexical variation within the discourse flow. We report two length-robust type-token measures. *MATTR* (Moving-Average Type-Token Ratio) averages the type-token ratio over a sliding fixed-length window, removing the well-known dependence of raw TTR on text length (Covington and McFall, 2010). *MTLD* (Measure of Textual Lexical Diversity) is the mean length of successive word sequences that maintain a type-token ratio above a fixed factor threshold (0.72), and is likewise designed to be stable across text lengths (McCarthy and Jarvis, 2010). The *Hapax ratio* is the proportion of types occurring exactly once (hapax legomena), a direct index of vocabulary richness and productivity (Baayen, 2001). *Zipf's* α is the exponent of the rank-frequency distribution, estimated from the log-log fit restricted to types with frequency ≥ 2 (Evert, 2004); higher values indicate a distribution more concentrated on a few high-frequency terms. The average internal cosine similarity *Avg. cos.* is the mean pairwise cosine between chunk vectors in a vector-space representation (Salton et al., 1975), used here as a measure of intra-corpus semantic cohesion (higher values indicating more homogeneous, less thematically varied text).

3.2.3 Information-Theoretic

All information theoretic measures are computed over the word-level distributions of each corpus group (in bits, $\log_2$), excluding placeholder tokens. Shannon entropy *Entropy* quantifies the uncertainty of the unigram word distribution, with higher values indicating a broader, more evenly used vocabulary (Shannon, 1948). Joint entropy *Joint H* extends this to contiguous word n-grams (2- and 3-gram), measuring the uncertainty of short word sequences. Conditional entropy *Cond. H* is the residual uncertainty of a word given its predecessors, so lower values indicate more predictable local phrasing. Global mutual information *Global MI* is the average mutual information between adjacent words, where higher values mean one word constrains the next more strongly on average (Manning and Schütze, 1999). *Mean PMI* is the mean pointwise mutual information over observed bigrams, capturing the strength of specific collocational bonds independently of their frequency (Church and Hanks, 1990; Manning and Schütze, 1999). Because Global MI is a frequency-weighted expectation (dominated by frequent, often function-word bigrams) whereas Mean PMI averages uniformly over bigram types, the two can diverge: the former reflects overall sequential predictability, the latter the strength of specialized collocations.

3.2.4 Syntactic and Morphological

Morphosyntactic features were computed from part-of-speech tags in the STTS tagset (Schiller et al., 1999). Metrics marked "(NL)" are restricted to natural-language tokens, excluding mathematical, unit, digit, and OCR-noise placeholders from both numerator and denominator. Nominal lexical density (*Nom. lex. den.*) is the proportion of noun tokens (STTS `NN`, `NE`) among lexical tokens, a standard index of the informational, nominal style typical of written and scientific registers (Biber, 1988). The *noun-verb ratio* (nouns to lexical verbs `VV*`, excluding auxiliaries and modals) likewise contrasts nominal against more verbal, narrative styles (Biber, 1988). Average noun-phrase length *NP length (NL)* approximates phrasal complexity by counting pre-nominal modifiers (`ART`, `ADJA`, `PDAT`, ...) preceding a nominal head, following the identification of dense phrasal noun-modification as the hallmark of academic complexity (Biber and Gray, 2016). Preposition density (*Preposition dens*) (proportion of prepositions) is a further marker of phrasal, informational packaging (Biber and Gray, 2010). Lexical density per sentence (*Lex. dens/sent*), the proportion of content words, follows Ure (1971) and Halliday (1985). Connective frequency *Connective freq.* indexes explicit cohesion through discourse connectives (Halliday and Hasan, 1976). *POS entropy* is the Shannon entropy of the POS-tag distribution (Shannon, 1948), with lower values indicating more predictable, formulaic tag sequences. The *Compound (NL)* proxy is the fraction of alphabetic tokens longer than twelve characters, a surface heuristic for German nominal compounds. Finally, verb-prefix (*Verb pref. (NL)*) distance is the mean number of tokens separating a finite verb from its separable prefix (the German verbal bracket, or *Satzklammer*); larger distances imply longer syntactic dependencies and greater processing demand (Gibson, 2000).

3.2.5 Cross-Corpus Distance

All distances are computed between each corpus group and the Leipzig baseline; higher values indicate greater divergence from general language. *Hellinger* distance and Jensen-Shannon (*JS divergence*) divergence both compare the two unigram probability distributions: the Hellinger distance is a bounded metric on $[0, 1]$ defined over the square-rooted probabilities (Cha, 2007), whereas JS divergence is the symmetric, smoothed form of the Kullback-Leibler divergence and, computed in bits, is likewise bounded in $[0, 1]$ (Lin, 1991). The Kolmogorov-Smirnov (KS) statistic is the maximum gap between two empirical cumulative distributions, applied here to sentence-length distributions (*KS sent. length*) (Massey Jr., 1951). Centroid distance is a metric from Latent Semantic Analysis (*LSA centroid*) and is defined as the cosine distance between corpus centroids in a shared latent-semantic space (TF-IDF weighting followed by truncated SVD; Deerwester et al., 1990), capturing latent topical structure rather than surface word frequencies. The out-of-vocabulary (*OOV rate*) rate is the percentage of a corpus's tokens absent from the Leipzig vocabulary. Together these measures follow the corpus-comparison tradition of quantifying how far a specialized corpus departs from a general-language reference (Kilgarriff, 2001).

4 Results

The examples in this section were selected manually and illustrate general claims made based on the metrics results.

4.1 Surface and Readability

Table 2: Surface & Readability metrics per corpus group. Cells are shaded per row from the row minimum (lightest) to the row maximum (darkest). Note that a higher F-Amstad score indicates an *easier* text, whereas for the remaining metrics higher values indicate longer or denser surface forms.

Metric	Leipzig	TeCoPhy	Mid.	High	Univ.	Para.	Wiki
F-Amstad (↑)	64.04	56.33	66.19	59.39	55.41	53.21	45.88
WSTF	8.407	9.817	8.457	9.613	10.03	10.33	11.63
Sent.(mean)	14.52	19.98	16.15	18.89	19.84	22.79	20.83
Sent.(std)	7.977	10.02	7.432	8.643	10.30	12.52	10.93
Word avg.	6.092	6.129	5.858	6.060	6.185	6.113	6.624
SymDens	0.041	0.084	0.063	0.083	0.109	0.043	0.049

As Table 2 shows, according to the readability formulas, Wikipedia is consistently the least readable, having the longest words and long sentences. Its relatively low symbol density within TeCoPhy may be indirectly related; by using less mathematical formalism, more of the conceptual load falls on the text, which becomes longer and more compound-laden (Example 1). Middle school is the most readable among physics subcorpora, considering the values from both readability formulas and the sentence and word lengths.

In general, there is a marked reduction in readability as the analyzed school levels progress, with a sharp drop between middle and high school. Symbol density increases consistently across school levels. Example 2 illustrates a high-symbol-density University sentence, where the physical relation is carried almost entirely by formalism rather than prose. Paradidactic texts are characterized by long sentences that impact readability, combined with a very low symbol density.

Overall, the physics categories are more demanding than the reference corpus, though this is driven more by longer sentences than by word length.

(1) *Hierfür wurden Graphitstreifen- und Phosphorstreifenaufdrucke auf Briefmarken und Fluoreszenzstreifen neben Ganzsachen-Wertzeicheneindrucke und phosphoreszierende Aufdrucke verwendet.* (Wikipedia)
'For this purpose, graphite-strip and phosphor-strip overprints on stamps and fluorescent strips were used, alongside postal-stationery value-mark imprints and phosphorescent overprints.'

(2) *Wegen der Energieerhaltung $\Delta W = \Delta E$ Oberfläche folgt $F \cdot \Delta x = \gamma \cdot 2l \cdot \Delta x$ oder $\gamma = F/2l$.* (University)
'Because of the conservation of energy, $\Delta W = \Delta E$ surface, it follows that $F \cdot \Delta x = \gamma \cdot 2l \cdot \Delta x$, or $\gamma = F/2l$.'

4.2 Lexical Diversity

As expected, the Leipzig Corpus presents the highest values for lexical diversity across global metrics such as MATTR, MTLD and Hapax ratio, reflecting its broad, general-language composition. Local metrics were not computed for the Leipzig Corpus since its texts cannot be fully reconstructed. Within the physics domain (TeCoPhy), paradidactic texts present the highest global and local lexical diversity. Moreover, it is interesting to observe the progression among the school levels: Middle school exhibits the highest thematic diversity (global MTLD), which likely reflects the pedagogical demand to contextualize physics concepts within a wide range of everyday themes. However, middle school also exhibits the highest Zipf's α (1.291) and the lowest Hapax ratio (0.345), indicating that despite the thematic variety, the core vocabulary remains more concentrated and repetitive. High school demonstrates a distinct dip in lexical diversity. Among the school levels, high

Table 3: Lexical Diversity metrics per corpus group with per-row heat map coloring.

Metric	Leipzig	TeCoPhy	Mid.	High	Univ.	Para.	Wiki
MATTR	0.835	0.819	0.823	0.814	0.818	0.828	0.813
MATTR (ord)	-	0.764	0.757	0.759	0.765	0.786	0.760
MTLD	171.1	134.9	133.0	122.1	131.9	146.0	123.4
MTLD (ord)	-	68.00	61.67	62.89	67.65	79.61	61.92
Hapax ratio	0.438	0.388	0.345	0.360	0.382	0.394	0.375
Zipf alpha	1.135	1.226	1.291	1.267	1.238	1.221	1.246
Avg. cos.	0.124	0.157	0.154	0.162	0.161	0.161	0.163

Table 4: Information-Theoretic metrics per corpus group with per-row heat map coloring.

Metric	Leipzig	TeCoPhy	Mid.	High	Univ.	Para.	Wiki
Entropy	8.518	8.247	8.192	8.189	8.241	8.296	8.200
Joint H (2-g)	10.69	11.04	10.77	10.93	11.01	11.27	11.06
Joint H (3-g)	10.82	11.35	11.04	11.21	11.31	11.61	11.35
Cond. H	2.450	2.749	2.690	2.752	2.735	2.808	2.774
Global MI	6.419	5.877	5.897	5.744	5.876	5.990	5.829
Mean PMI	2.109	2.176	2.297	2.478	2.214	2.025	2.382

school exhibits the lowest global MTLD value; together with the high average cosine similarity, this indicates that high school physics texts are more standardized and less thematically varied. University sits close to middle school with a high global MTLD but likely reflecting the diversity of sources and subfields the subcategory spans.

The vocabulary-concentration metrics — Zipf's α and the Hapax ratio — show a monotonic progression from Middle through High School to University. This is consistent with a gradual expansion of the technical vocabulary, even as the thematic scope temporarily narrows during the high-school years. More generally, Zipf's α indicates that the frequency distribution of physics texts is heavily concentrated on high-frequency terms, whereas the Leipzig Corpus features a much more diverse distribution. The higher average cosine similarity in the physics categories is further consistent with this tighter semantic and lexical cohesion.

4.3 Information-Theoretic

The information-theoretic metrics clearly distinguish the domain-specialized physics corpus (TeCoPhy) from the Leipzig Corpus. The latter exhibits a broader general vocabulary (highest entropy), but relies on more formulaic and predictable phrasing (lowest conditional and joint entropies and highest Global MI). In contrast, the physics corpora use a more restricted vocabulary (lower entropy) but show higher joint and conditional entropies, indicating that word sequences are less predictable from immediate context than Leipzig's. This combination is what a narrow vocabulary recurring across many different local contexts would produce. Mean PMI — highest at High-school and Wikipedia — reflects the tightly bound technical jargon and collocations on which they rely; Global MI and Mean PMI thus order the groups differently, the former capturing general-language predictability, the latter the strength of specialized collocations. Paradidactic texts are the exception within the domain: they show the highest joint and conditional entropies, indicating lower predictability and a greater variety of phrasal constructions, consistent with a more informal and creative style.

4.4 Syntactic and Morphological

The noun-verb ratio is lowest in Middle school (2.551), indicating a more narrative, action-oriented style for explaining physical phenomena (Example 3). Both the noun-verb ratio and nominal lexical density identify High school as the most nominalized among the school levels, supporting our hypothesis of a representational shift at university toward mathematical symbolism, whereas High-school texts must still convey technical, formal content in natural language (Example 4). Wikipedia and the Leipzig Corpus are the most nominalized overall (high noun-verb ratio and nominal density), reflecting their informational, non-narrative function.

Table 5: Syntactic & Morphological metrics per corpus group with per-row heat map coloring.

Metric	Leipzig	TeCoPhy	Mid.	High	Univ.	Para.	Wiki
Nom. lex. den.	0.252	0.229	0.230	0.236	0.227	0.226	0.245
NP length (NL)	1.692	1.852	1.782	1.841	1.872	1.806	1.832
Noun-verb ratio	3.095	2.885	2.551	2.924	2.875	2.850	3.498
POS entropy	4.316	4.296	4.258	4.236	4.296	4.379	4.137
Preposition dens	0.098	0.095	0.093	0.096	0.098	0.093	0.106
Compound (NL)	0.070	0.075	0.056	0.066	0.077	0.068	0.099
Connective freq.	3.982	3.936	4.156	3.737	4.020	4.732	4.247
Lex. dens/sent.	0.466	0.454	0.455	0.458	0.448	0.454	0.476
Verb pref. (NL)	6.164	7.080	6.151	6.780	6.514	6.544	7.229

Physics texts show greater phrasal elaboration: average noun-phrase length is higher across the physics subcorpora than in Leipzig (TeCoPhy 1.852 vs. 1.692), and the University corpus reaches the highest NP length among the educational levels (1.872), reflecting the need to embed multiple modifiers into single conceptual units for scientific precision. Compounding shows a different pattern: the school-level physics texts compound less than Leipzig (Middle 0.056, High 0.066 vs. Leipzig 0.070), and only University (0.077) and, most strikingly, Wikipedia (0.099) exceed it. Wikipedia's

high compound proxy corroborates its poor readability, its texts being burdened by long single-word compounds rather than by mathematical symbols. The verb-prefix distance is on average larger in physics than in Leipzig, with Middle school the lowest of all groups, although this metric varies considerably across draws. Paradidactic texts combine the most usage of connectives (4.732), consistent with an explanatory style built on explicit transitions (Example 5), whereas high school uses the fewest (3.737) among physics subcorpora; together with its low POS entropy, this supports the hypothesis that high-school texts are more standardized and formulaic. Finally, the lower nominal density of physics, set against its longer noun phrases, suggests that physics nouns rarely appear in isolation but are instead surrounded by modifiers.

(3) *Wenn du es loslässt, beginnt es sich zu drehen und sinkt herab, bis der Faden abgewickelt ist.* (Middle School)
'When you let it go, it starts to spin and descends until the string has unwound.'

(4) *Werden die parallelen Kondensatorplatten durch andere oder durch ein beliebig geformtes Leiterpaar ersetzt, so ergibt sich auch in diesen Fällen, dass die Ladung der angelegten Spannung proportional ist; die Kapazität C hängt dann jedoch in anderer Weise von der Form und der geometrischen Anordnung des Leiterpaares ab.* (High School)
'If the parallel capacitor plates are replaced by other plates, or by an arbitrarily shaped pair of conductors, it likewise follows in these cases that the charge is proportional to the applied voltage; the capacitance C, however, then depends in a different way on the shape and geometric arrangement of the pair of conductors.'

(5) *Die ganzen neuen Teilchen sind deshalb nicht "nützlich", weil sie entweder in Sekundenbruchteilen in andere Teilchen zerfallen oder weil sie kaum zu "fassen" sind, da sie nur extrem schwach mit uns wechselwirken, also mit der Materie, aus der wir und unsere Umwelt bestehen.* (Paradidactics)
'All these new particles are therefore not 'useful', because they either decay into other particles within fractions of a second, or because they are hardly to be 'grasped' at all, since they interact only extremely weakly with us — that is, with the matter of which we and our environment consist.'

4.5 Cross-Corpus Distance

These metrics indicate that physics language grows increasingly distant from general language as the school level advances, as the OOV rate indicates more clearly (4.46 → 5.64 → 7.21 %). Hellinger distance (0.657 → 0.660 → 0.665) and JS divergence (0.493 → 0.497 → 0.502) order the levels the same way, though the steps between adjacent levels are small. The rising OOV rate plausibly tracks the compound proxy (Table 5), which follows the same ordering, reflecting technical jargon and compounds absent from the Leipzig Corpus. Wikipedia and University are the two most distant subcorpora from Leipzig on Hellinger, JS, and OOV alike.

Table 6: Cross-corpus distance metrics relative to the Leipzig baseline (Leipzig itself omitted, as it is the reference) with per-row heat map coloring. Higher values indicate greater distributional distance.

Metric	TeCoPhy	Mid.	High	Univ.	Para.	Wiki
Hellinger	0.658	0.657	0.660	0.665	0.637	0.669
JS divergence	0.494	0.493	0.497	0.502	0.478	0.509
KS sent. length	0.293	0.191	0.298	0.289	0.365	0.324
LSA centroid	0.141	0.160	0.154	0.145	0.099	0.167
OOV rate (%)	6.525	4.459	5.636	7.211	4.192	6.777

Paradidactic texts present a revealing exception. Their vocabulary and themes are the closest of all subcorpora to Leipzig — the lowest values for Hellinger distance (0.637), JS divergence (0.478), LSA centroid distance (0.099), and OOV rate (4.19 %) — plausibly reflecting a reliance on accessible, everyday topics. Yet at the sentence level they diverge most sharply: paradidactic texts show the highest KS sentence-length statistic (0.365), consistent with long, loosely structured explanatory sentences. Middle-school texts, by contrast, have the lowest KS value (0.191), most closely mimicking Leipzig's sentence-length distribution.

The one metric that does not follow the school-level progression is the LSA centroid distance ranking Middle school (0.160) as the most distant, with High School (0.154) and University (0.145) closer to Leipzig but with values too close to separate. This apparent inversion may be an artifact of the University corpus's greater internal thematic heterogeneity, which pulls its centroid toward the center of the semantic space and thus closer to Leipzig, even as its surface vocabulary is the most divergent.

5 Discussion

Taken together, the metrics support a compact profile of the corpora and answer which metrics most sharply separate physics from general language (RQ1). The strongest discriminators between physics and the Leipzig Corpus are noun-phrase length, average cosine similarity and symbol density, capturing respectively the heavy modification of physics nouns, its internal thematic cohesion and its reliance on mathematical formalism. Within the physics subcorpora, each register carries a distinctive signature. University is set apart above all by its symbol density, the highest of any group. High school is marked by dense nominalization and

the strongest collocational binding, together with the lowest connective frequency. Middle school is the mirror image: short, simple sentences with the most verbal, action-oriented style. Paradidactic texts combine the highest connective frequency with high POS entropy — an explanatory, loosely structured register. Wikipedia stands out for its long single-word compounds and high lexical density. This profile shows that no single family suffices: lexical, syntactic, information-theoretic, and distributional metrics each isolate a different axis on which physics language departs from general written German.

The most significant pattern runs counter to the intuition that academic language is universally *denser*: Leipzig actually exceeds physics on MTLD and nominal density. The resolution lies in the distinction between diversity and concentration: general language draws on a broad vocabulary to range across topics, whereas physics concentrates on a narrow terminological core that recurs with high precision. Technical language is thus not simply *more complex* than general language; it is structurally closed and cohesive, indexed by its high OOV distance and internal semantic cohesion (average cosine similarity). Consistent with Fraser (2025), this suggests that surface simplification of physics texts may fail when the difficulty lies in terminological closure rather than in surface form. Moreover, although nominal density is higher in Leipzig, its noun phrases are shorter, indicating more numerous and less modified nominal heads. Physics, in contrast, uses fewer heads carrying more modification, which is plausibly more demanding to process.

The progression across educational levels is non-monotonic and appears to be driven by representational shifts. The standard account of academic German posits increasing nominal style as texts advance; our data instead show nominal density peaking at high school (0.236) and dropping at university (0.227). We interpret this as a trade-off: at high school, natural language carries the entire explanatory load for abstract concepts, whereas at university part of this structural work is offloaded to mathematical formalism. The apparent simplicity of university texts on syntactic complexity metrics is therefore partly an artifact of this shift: university prose requires fluent parsing of an additional semiotic system to which surface readability formulas are structurally blind. Within this progression, the high-school corpus represents a linguistic *bottleneck*. Among the school levels it exhibits the highest nominal density and the highest Mean PMI, alongside a sharp readability drop from middle school. We hypothesize that, lacking the mathematical resources available at university, high-school texts are syntactically elaborated to preserve precision. The paradidactic register solves the same communicative problem by opposite means: although its vocabulary is the closest to Leipzig, it relies heavily on connectives to scaffold logical relationships — a strategy essentially inverse to that of high-school texts, which minimize connectives.

In general, the evaluated metrics fail to converge on a single ordering; instead, their divergences highlight how each defines the locus of difficulty (RQ2). For example, Leipzig surpasses physics in both MTLD and nominal density, indicating that lexical diversity metrics overlook terminological concentration. Readability formulas and distributional distances similarly place the paradidactic corpus at opposite extremes. Despite being the closest subcorpus to Leipzig in vocabulary and topic, F-Amstad rates it significantly harder (53.21 vs. 64.04) solely because its sentences are longer. Furthermore, the LSA centroid and OOV rate conflict regarding which school level most closely mirrors general language, as LSA captures average position in a semantic space, while OOV measures word presence.

These divergences have direct implications for automated text simplification of technical texts, which relies on these metrics to diagnose difficulty and measure intervention. If physics texts are difficult due to terminological closure rather than surface complexity, a system outputting a paradidactic profile — utilizing everyday vocabulary, connective scaffolding, and consequently longer sentences — would be incorrectly penalized by standard formulas. In practice, this profile inverts automated simplification: paradidactic texts reduce lexical distance and insert connectives while *lengthening* sentences, directly countering the standard operations of sentence splitting and connective reduction. Although this paradidactic corpus is a distinct genre rather than a parallel simplification, it is a register written for a broader audience. Its profile strongly suggests that standard surface-level simplification tools and metrics are ill-equipped to handle this specific dimension of textual difficulty.

6 Conclusion

This study characterized German physics-teaching language across five text categories using 33 quantitative metrics. Our findings indicate that technical language constitutes a structurally distinct register, and that the interplay between natural language and mathematical formalism is central to how its complexity and readability should be measured. Three specific patterns illustrate this: the *high-school bottleneck*, the *university offloading* effect, and the distinctive discourse of paradidactic texts. Our interpretation repeatedly invokes a dimension our surface metrics capture only indirectly: the abstractness of the concepts a text encodes. A natural next step is to profile the corpora along semantic dimensions that surface features leave implicit — lexical abstractness and concreteness, conceptual hierarchy, and figurative or ambiguous usage — to test directly whether the *explanatory load* we attribute to high-school syntax corresponds to measurably greater abstractness. Combined with reader comprehension data and metrics that model the interplay between natural language and mathematical notation, this would move from characterizing the surface of physics-teaching language toward explaining its conceptual difficulty.

Limitations

This study is largely descriptive and exploratory, and several limitations bound its conclusions. Many metrics were estimated from ten resampled draws per corpus category, which is too few for stable interval estimations. Inspecting the draws, the separations that hold across all or nearly all draws include symbol density and noun-phrase length between physics and the reference corpus, the OOV rate across school levels, and sentence length for the paradidactic subcorpus. Verb-prefix distance, Hellinger distance and JS divergence do not separate consistently. Claims resting on these should be read as descriptive of the samples rather than established differences. Four further metrics (MATTR, MTLD, Zipf's α, and average cosine similarity) were computed on a single sample truncated to the smallest category (36,034 sentences; 778,943 tokens for the token-level measures) and carry no dispersion estimate, although their larger sample size makes them correspondingly more stable. The mathematical formalism in TeCoPhy is captured only partially: OCR and pre-processing filter or gate out a portion of the mathematical content, so symbol density understates the true mathematical load, and the representational-shift effects we report should be read as a lower bound. Additionally, the Leipzig Corpus is not available as continuous texts, which precludes sequential and document-level metrics for it (hence the absent *(ord)* values). Finally, the category comparisons are cross-sectional: we do not track content across categories, so developmental readings such as the offloading interpretation describe associations, not demonstrated processes.

Lay Summary

A text can be difficult in various ways, and we wanted to understand what makes a physics text particularly challenging in German. Sometimes a text is difficult because of long sentences or long words. But physics texts are often difficult because they differ greatly from what we encounter in everyday reading. This study analyzes the difficulty and complexity of physics texts from various sources: middle-school, high-school and university textbooks, Wikipedia articles, and popular-science books. We measured 33 characteristics of these texts and compared the results with texts from news sites and other online sources.

We found that physics texts are not more difficult in every respect. They actually use a smaller variety of words than everyday texts. What makes them hard is that many of their words never appear in everyday German at all, and that these words are packed into long, dense constructions. The difficulty also changes with school level. In high-school texts, abstract concepts must often be explained in words alone, and the language becomes especially dense. At university, mathematics takes over much of this work: the wording carries less of the load, and the difficulty shifts toward reading the formulas.

Popular physics texts, written for a general audience, take a different approach. They use vocabulary close to everyday language and add many linking words, such as *because* and *therefore*, to make the reasoning explicit. This makes their sentences longer. Because standard readability formulas look mainly at sentence length, they rate these texts as considerably harder than everyday German, even though their vocabulary is the closest to it of all the physics texts we studied. This matters for simplification strategies. For example, to shorten sentences and remove linking words is the opposite

of the strategy human writers use when explaining physics to a general audience.

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